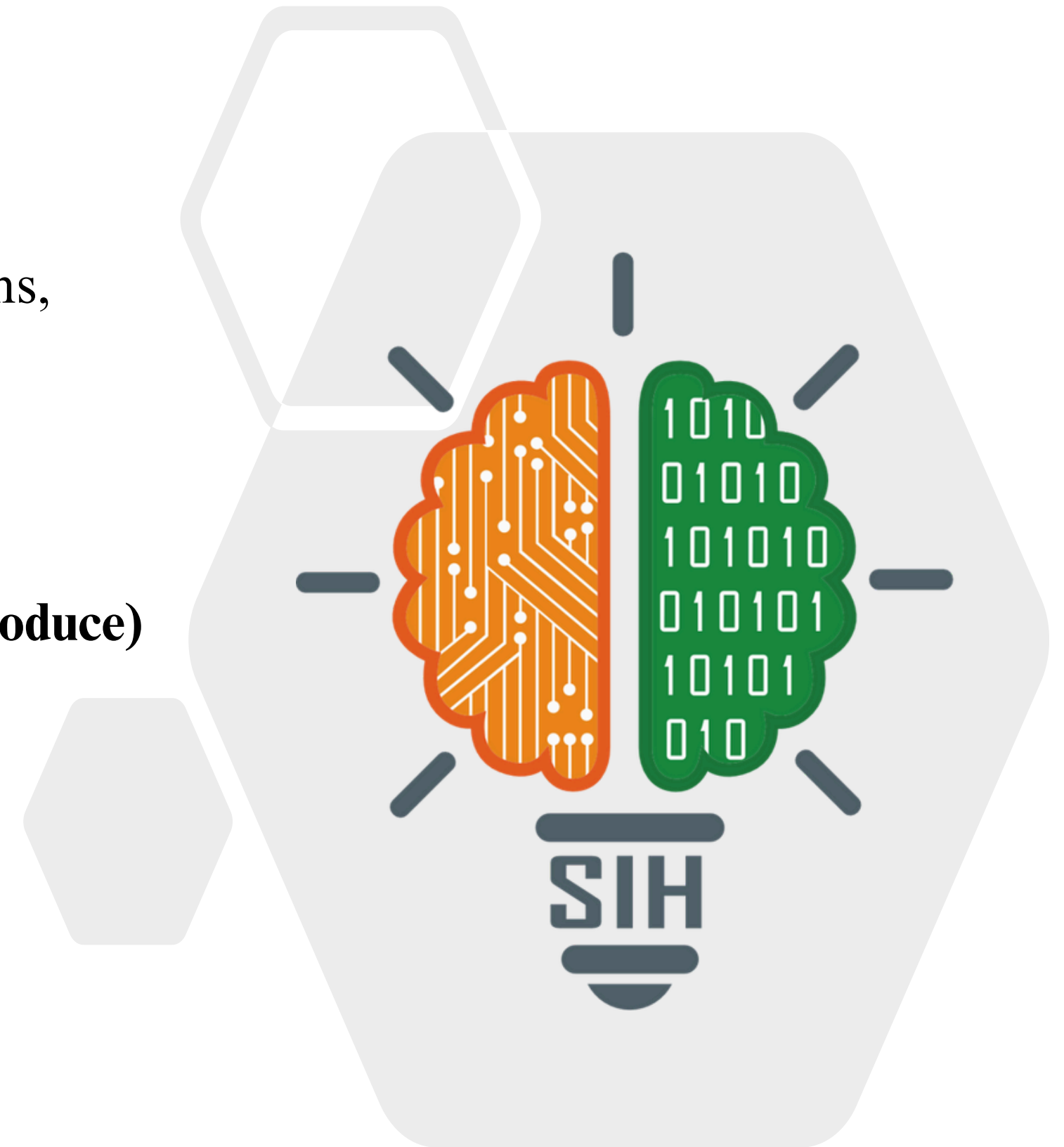


SMART INDIA HACKATHON 2026



- **Problem Statement ID: SIH26193**
- **Problem Statement Title:** Student Innovation-Developing solutions, keeping in mind the need to enhance the primary sector of India - Agriculture and to manage and process our agriculture produce.
(Blockchain-Based Supply Chain Transparency for Agricultural Produce)
- **Theme** : Agriculture, FoodTech & Rural Development
- **PS Category:** Software
- **Team Name:** 3LPA
- **Team ID:** 133905



SOLUTION

Seed2Shelf combines Blockchain, QR traceability, and Smart Contracts to build a transparent and trusted farm-to-shelf supply chain.

- **Blockchain Traceability** records every transaction and ownership handoff securely on blockchain.
- **QR-Based Verification** allows to scan a QR code to verify the complete journey of agricultural produce.
- **Unified Marketplace** connects farmers, processors, distributors, retailers, and consumers directly.
- **Verified Onboarding** ensures only verified stakeholders participate in the supply chain.
- **Smart Escrow** automatically releases secure payments when predefined conditions are fulfilled.
- **Digital Agriculture Ready** complements **AgriStack** and **e-NAM** with verifiable supply-chain traceability.

We make India’s agricultural supply chain more transparent, trusted, and accountable, creating a secure digital journey from Seed to Shelf.

(Seed2Shelf is **80% complete**, backed by **strong ground-level surveys**; testing and validation are currently ongoing.)



End-to-End Traceability: Unlike traditional mandi-level tracking, Seed2Shelf records the complete farm-to-shelf journey of produce on blockchain, making origin and ownership verifiable.



Direct Value & Price Transparency: Farmers, processors, distributors, retailers, and consumers are connected on one marketplace, with price and ownership history visible at every stage, reducing information asymmetry.



Trust Through Verification: Stakeholders are onboarded using existing Indian credentials such as **Aadhaar**, **land records**, **FSSAI**, **GST**, and **APEDA**, while blockchain provides tamper-resistant records.



Built for India’s Digital Agriculture Ecosystem: Seed2Shelf is designed to complement platforms such as **AgriStack** and **e-NAM**, adding the missing layer of verifiable supply-chain history rather than replacing existing infrastructure.

PROTOTYPE



TECHNICAL APPROACH

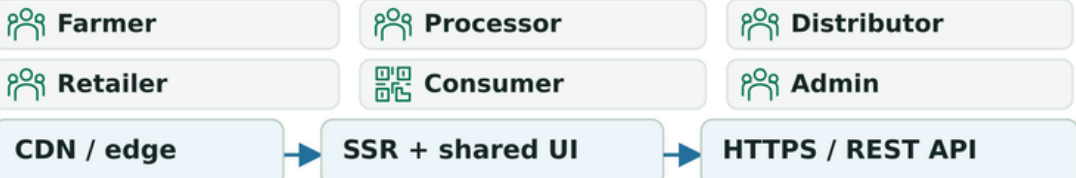
01 FRONTEND ARCHITECTURE

Six role-based Hubs • Shared components • Responsive web



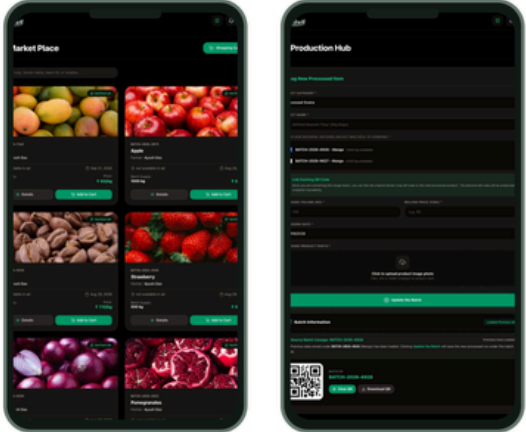
- Next.js**
SSR + routes
- React**
Reusable UI
- TypeScript**
Typed clients

YOUR SEED2SHELF INTERFACE



02 MOBILE ARCHITECTURE

Installable PWA • Lightweight screens • Assisted access



- PWA**
Web app manifest
- Service worker**
Caches the app shell and selected read views.
- IndexedDB local outbox**
Saves drafts on-device while disconnected. [2]
- Reconnect + retry**
Foreground sync; background sync where supported.



Regional-language UI + CSC/FPO support. **SMS fallback** needs mobile signal; it does not provide disconnected blockchain settlement.

QR lookup → public product history + verification status

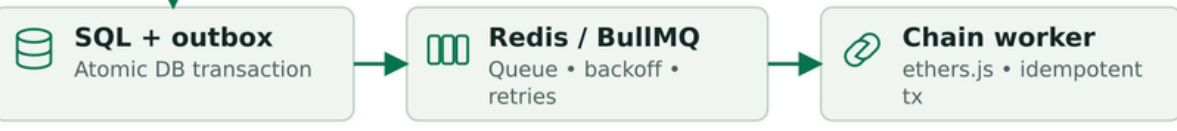
03 BACKEND ARCHITECTURE

- Node.js**
Runtime
- NestJS**
Domain API
- Redis**
Cache + queue
- BullMQ**
Async workers

API GATEWAY
HTTPS • authentication • RBAC • rate limits • request validation

- Auth + KYC**
Roles • approval status
- Orders + inventory**
Lots • quantities • pricing
- Escrow + payments**
Holds • release • refunds
- Wallets + signing**
User binding • key policy
- Shipments**
Dispatch • acceptance
- Notifications + analytics**
Alerts • reports • audit views

DURABLE WRITE + ASYNC COMMIT



RECEIPT LISTENER + RECONCILIATION
Confirmed events → update database projection → notify the user

Durable outbox prevents lost events; retries must be idempotent. [3, 4]

04 DATABASE ARCHITECTURE

- PostgreSQL**
Off-chain records
- Object storage**
Encrypted documents
- Redis**
Short-lived cache

- POSTGRESQL PRIMARY**
Users • KYC references • orders • invoices • shipment events
- PRIVATE OBJECT STORE**
Encrypted KYC files + evidence; access-controlled references



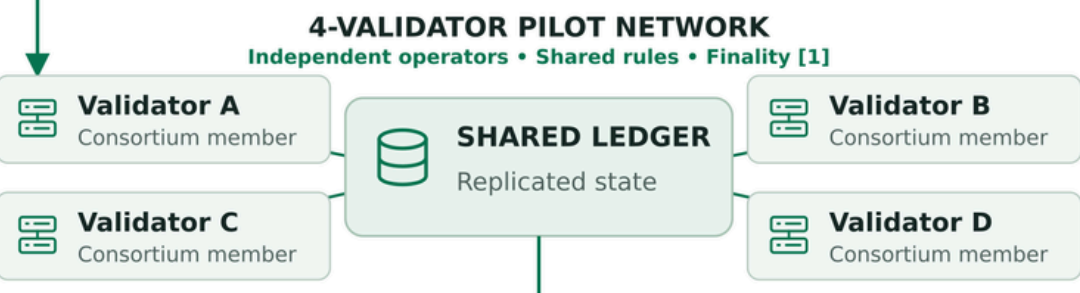
Scale on evidence: indexed queries first; add read replicas and region/time partitions when measured traffic needs them.

Recovery: backups + point-in-time restore. **On-chain:** compact commitments and contract state; keep personal documents off-chain.

05 BLOCKCHAIN ARCHITECTURE

- Besu**
Permissioned EVM
- Solidity**
Smart contracts
- QBFT**
Consensus [1]

PERMISSIONED JSON-RPC + SIGNER
Allowlisted access • isolated signing keys • authenticated submissions

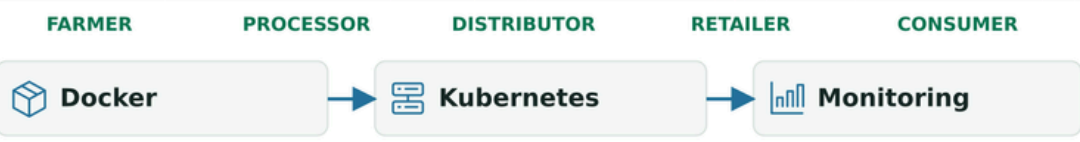


- IDENTITY + TRACEABILITY**
Wallet binding • batch commitments
- OWNERSHIP + ESCROW**
Authorized transfers • payment state



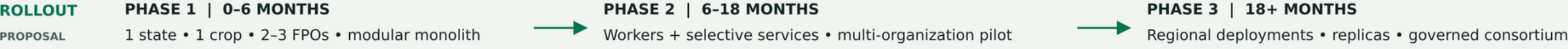
Reject / dispute → hold, refund or resolution under agreed rules.
Test-token escrow for the demo; INR settlement needs a payment-provider integration. Membership is proposed.

06 OPERATIONS & DEPLOYMENT



API Deployments + HPA: scale by demand. **Data services:** managed HA or StatefulSets with persistent volumes.

Security: encryption • least privilege • separate audit logs. **Planned adapters:** e-NAM / AgriStack / document verification.



Current baseline: supplied repo description.
Migration is guided by measured load.
No national capacity benchmark claimed.

FEASIBILITY



Technical Feasibility

Blockchain-based traceability, smart contracts, QR verification, and integrated APIs provide a feasible foundation for Seed2Shelf’s architecture.



Operational Feasibility

FPOs, digital farmer networks, eNAM, and logistics partners enable practical adoption across the supply chain.



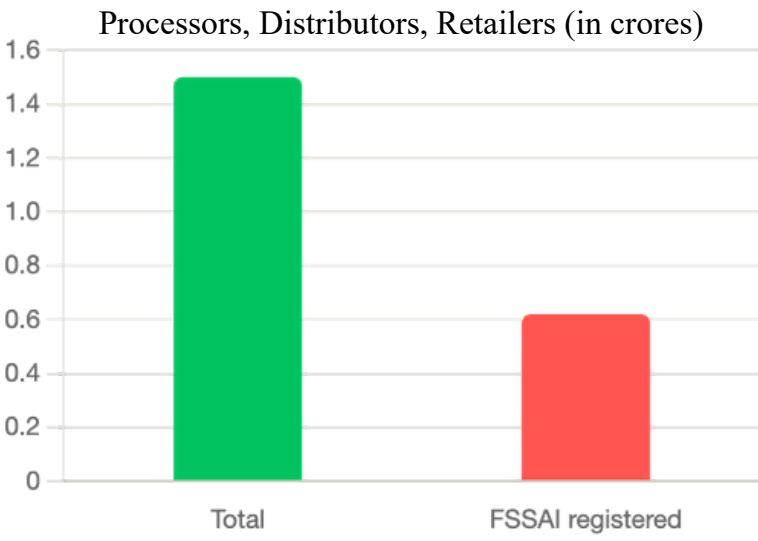
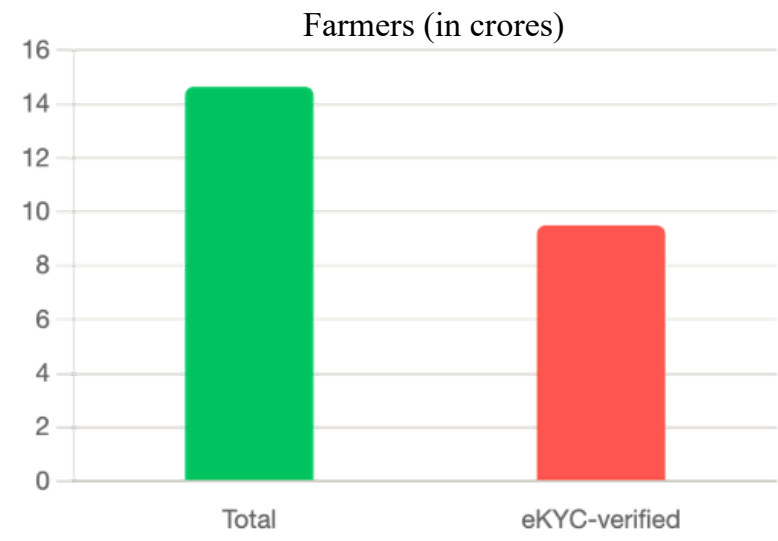
Economic Feasibility

Large agricultural market, transaction volumes, and multiple revenue streams make Seed2Shelf commercially viable.



Regulatory Feasibility

FSSAI, GST, APEDA, and AgriStack compatibility supports compliant implementation.



CHALLENGES



- 1. Low Network Reliability** – Rural areas have ~79% internet access, but reliable broadband remains limited.
- 2. Low Blockchain Literacy** – Blockchain/DLT concepts are unfamiliar to many farmers and FPO staff, creating adoption challenges.
- 3. National-Scale Complexity** – Real-time, tamper-proof ledger synchronization across 14.65 crore farmers + 1.5 crore supply-chain actors is highly challenging.



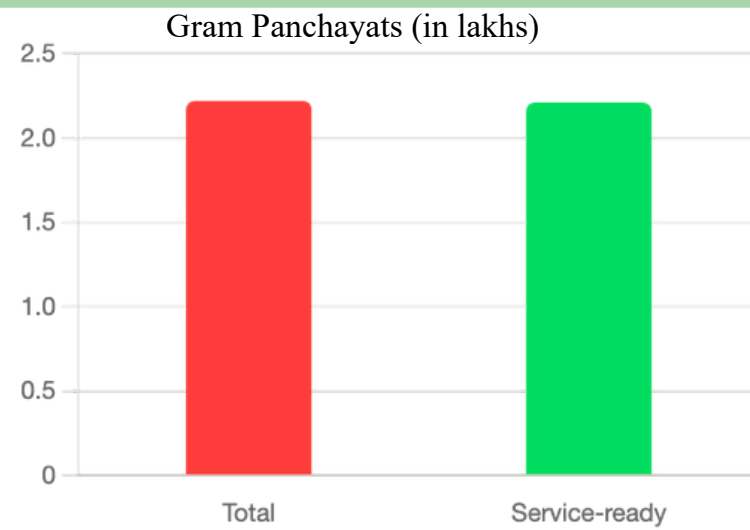
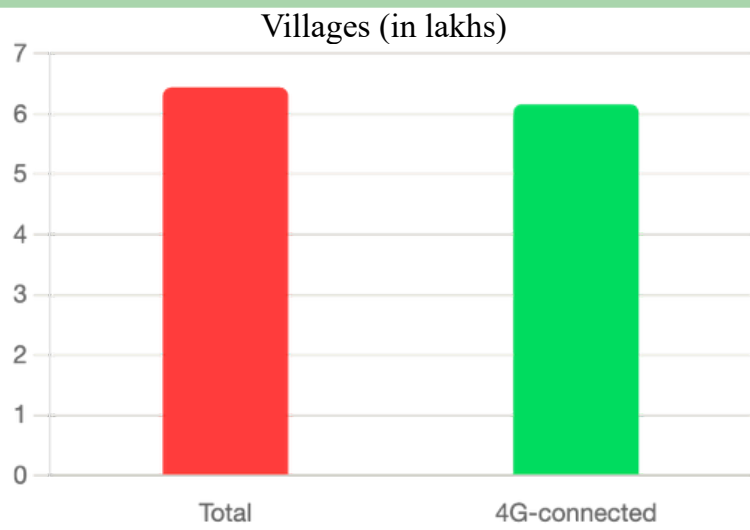
VIABILITY

Fast-growing digital base

- **9.5 crore farmers (65%)** are eKYC-verified; rural internet grew **59%→79%** in 4 years. **10,000 FPOs** already reach **56.3 lakh** farmers, with **~3,500 on eNAM** — a ready-made rollout channel.
- Pilot via existing FPOs, integrate with **eNAM/AgriStack**.
- Reach remaining users via **regional-language, offline/SMS** design.

Low incremental cost, given existing infrastructure

- DPI for Agriculture already covers 6 crore farmers' land records. FPO scheme funding (**₹6,865 Cr outlay, ₹600 Cr+** equity grants, **₹2,000 Cr** NABARD fund) shows this kind of digitization is already being financed.
- Leverage **AIF, PMFME, and Operation Greens** subsidies instead of new funding — keeping launch cost low.



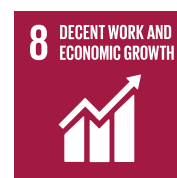
IMPACT AND BENEFITS

Economic Benefits

Fairer prices, reduced middlemen exploitation, lower wastage, and better market access strengthen farmer income and economic growth



SDGs:



Social Benefits

Greater transparency, food-safety assurance, digital inclusion, and farmer empowerment build trust, equity, and stronger rural communities



SDGs:



Environmental Benefits

Traceable logistics, reduced post-harvest waste, and real-time climate/supply monitoring enable more sustainable and climate-resilient agriculture



SDGs:



- Targets the **94% (≈13.8 crore)** currently outside **MSP protection** by giving every farmer a direct digital storefront, bypassing the **mandi/MSP** dependency entirely.
- Cuts the **60–70%** middlemen margin by enabling direct buyer-to-farmer sales with escrow, and removes the **25–30 km mandi-distance barrier** since listings happen from the farm.



- Brings the **~70%** unorganised processing base into a **verified digital sourcing chain**, replacing informal paper records with **blockchain-logged provenance**.
- Raises the **~41% FSSAI-digital-registration rate** by making compliance onboarding part of joining the platform, not a separate effort.



- Tackles the **₹1.53 lakh crore** annual **post-harvest loss** with real-time shipment tracking and **QR checkpoints** at every handoff.
- Makes the **97% / 91% / 85% cold-chain infrastructure gaps visible** and quantifiable for the first time, enabling targeted investment instead of blind losses.



- Brings the **89% of undigitised kirana stores** onto a lightweight, low-bandwidth inventory and sourcing app — **no POS hardware required**.
- Gives **tech-unexposed kiranas** a single onboarding path through **FPOs/CSCs** instead of adopting procurement and payment tools separately.



- Directly targets the **1-in-4 failed food-safety samples** by giving every product a **scannable, tamper-proof farm-to-shelf** journey.
- Rebuilds the trust gap (**84% distrust authorities, 68% distrust sellers**) by replacing word-of-trust with verifiable, on-chain proof at the point of purchase.

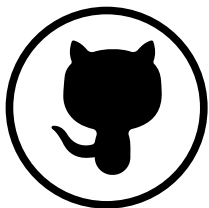
STAKEHOLDER BENEFITS

IEEE Research Publications

- [Blockchain-Based Agri-Food Supply Chain: A Complete Solution, 2020](#)
- [Smart Contract-Based Agricultural Food Supply Chain Traceability, 2021](#)
- [Blockchain-Based Soybean Traceability in Agricultural Supply Chain, 2019](#)
- [Agricultural Supply Chain Management System using Blockchain, 2025](#)
- [Blockchain Application for Sustainable Supply Chain Management in Indian Agriculture, 2025](#)

Government Policies

- [Digital Agriculture Mission: Tech for Transforming Farmers’ Lives](#)
- [PM Kisan Samman Nidhi Website](#)
- [ENAM Website](#)



Check out our Code Files Here : [Link](#)



Check out our Software UI Snapshots Here: [Link](#)



Check out our Prototype Software Demonstration Here: [Link](#)

Market Opportunity

Market Validation

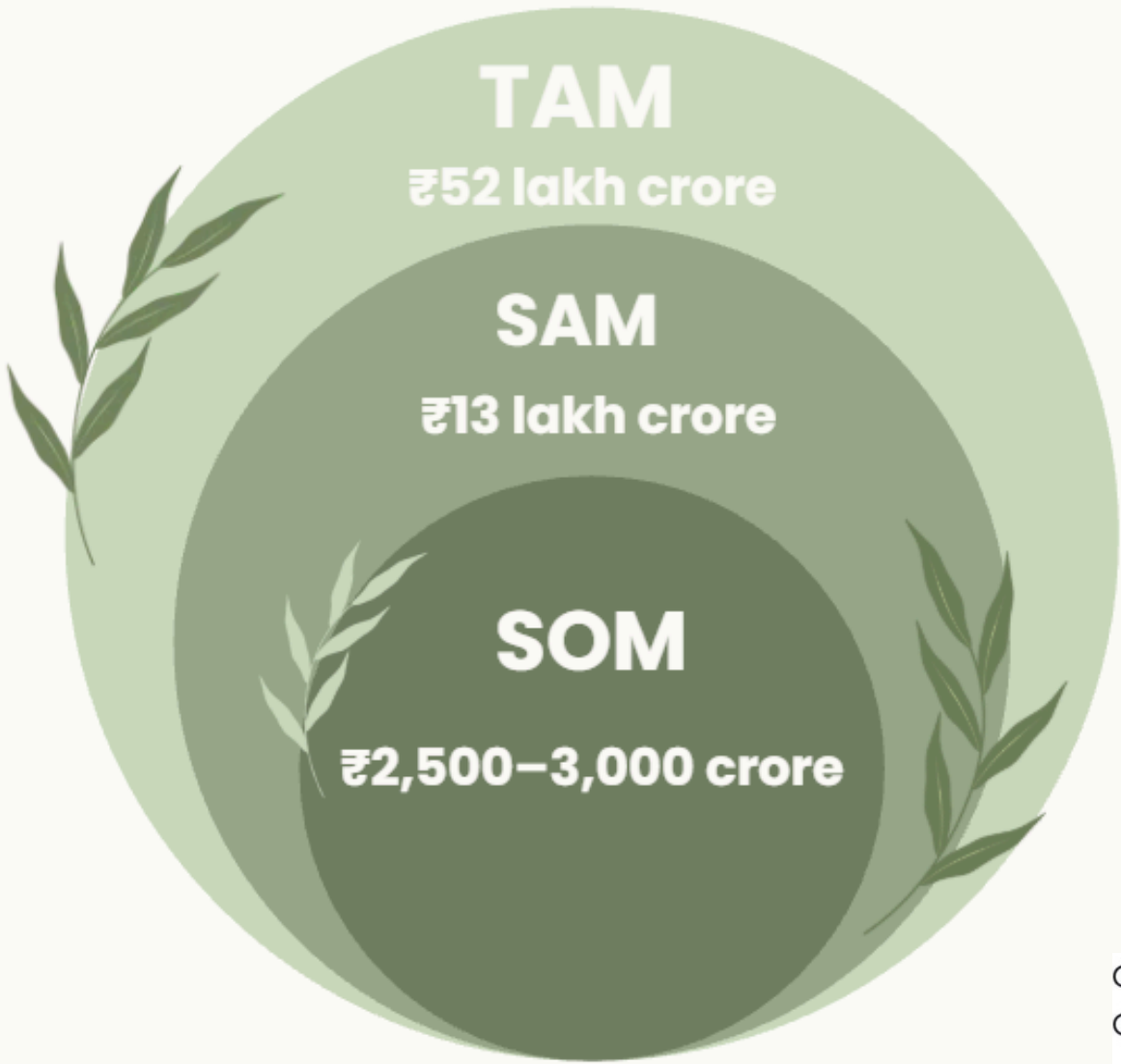
- ₹52.09 lakh crore agriculture GVA (2025–26)
- 55% of India's population depends on agriculture
- 86% of farmers are small & marginal (<2 ha)
- 10,000+ FPOs enable scalable onboarding

Digital Readiness

- e-NAM: ₹4.84 lakh crore cumulative trade
- 1.80 crore farmers, 2.73 lakh traders
- 1,656 connected mandis
- e-NAM 2.0 launched in 2026

Market Opportunity

- TAM: ₹52 lakh crore GMV
- SAM: ₹13 lakh crore GMV
- SOM (Yr 3–5): ₹2,500–3,000 crore GMV
- Revenue assumes a conservative 1% transaction fee



NATIONAL STRATEGY ON BLOCKCHAIN

Towards Enabling Trusted Digital Platforms



India’s National Strategy on Blockchain promoting Agriculture: [LINK](#)

Government of Odisha



Promoted by Govt. of Odisha in SIH 2025 Problem Statements.